Take Home Activity 2

Student Number – 35040

Q1

```
// Q1
import java.util.Scanner;

public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Enter the First Subject Mark: ");
        double sub1 = scanner.nextDouble();

        System.out.print(s: "Enter the Second Subject Mark: ");
        double sub2 = scanner.nextDouble();

        System.out.print(s: "Enter the Third Subject Mark: ");
        double sub3 = scanner.nextDouble();

        double total = (sub1 + sub2 + sub3);
        double average = (total / 3);
        double percentage = (total / 300) * 100;

        String grade;
        if (percentage >= 75) {
            grade = "Distinction";
        } else if (percentage >= 60) {
            grade = "Credit";
        } else if (percentage >= 50) {
            grade = "Pass";
        } else {
            grade = "Fail";
        }

        System.out.println("Total: " + total);
        System.out.println("Average: " + average);
        System.out.println("Percentage" + percentage);
        System.out.println("Grade: " + grade);


        scanner.close();
    }

}
```

```
Enter the First Subject Mark: 75
Enter the Second Subject Mark: 80
Enter the Third Subject Mark: 95
Total: 250.0
Average: 83.33333333333333
Percentage83.33333333333334
Grade: Distinction
```

Q2

```java
// Q2
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
    int x = 5;
    int y = ++x + x++ + --x + x--;

    System.out.println("x = " + x + ", y = " + y);

    }
}
```

```
x = 5, y = 24
```

- Pre-increment changes value before using it, post-increment uses value before changing it.

Q3

```java
// Q3
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        int a = 10;
        int b = 5;

        System.out.println(a > 2 && b > 12);

        System.out.println(a > 2 || b > 12);

        System.out.println(!(a > 2 && b > 12));
    }

}
```

```
false
true
true
PS D:\NSBM\1 Year\2 Sem\Java
```

Q4

```java
// Q4
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        int num = 5;

        if (num > 0) {
            System.out.println(x: "Number is Positive");
        }
    }
}
```

```
ming\Code\User\workspaceSto
Number is Positive
PS D:\NSBM\1 Year\2 Sem\Jav
```

Q5

```java
// Q5
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        int a = 5;

        if (a % 2 == 0) {
            System.out.println(x: "Number is Even: ");
        } else {
            System.out.println(x: "Number is Odd: ");
        }
    }
}
```

```
ming\Code\User\workspa
Number is Odd:
PS D:\NSBM\1 Year\2 Se
```

Q6

```java
// Q6
import java.util.Scanner;

public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Enter the BMI value: ");
        double BMI = scanner.nextDouble();

        String bmi_value;
        if (BMI < 18.5) {
            bmi_value = "Underweight";
        } else if (BMI <= 24.9) {
            bmi_value = "Normal";
        } else if (BMI <= 29.9) {
            bmi_value = "Overweight";
        } else {
           bmi_value = "Obese";
        }

        System.out.println("A Person's BMI value: " + bmi_value);

    scanner.close();
    }
}
```

```
ming\Code\User\workspaceStorage\f/
Enter the BMI value: 30
A Person's BMI value: Obese
```

Q7

```java
// Q7
import java.util.Scanner;

public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print(s: "Enter the First Number: ");
        double num1 = scanner.nextDouble();

        System.out.print(s: "Enter the Second Number: ");
        double num2 = scanner.nextDouble();

        System.out.print(s: "Enter the Operator (+ , - , * , / , %): ");
        char op = scanner.next().charAt(index: 0);

        double result;
        switch (op) {
            case '+':
                result = num1 + num2;
                System.out.println("Result: " + result);
                break;

            case '-':
                result = num1 - num2;
                System.out.println("Result: " + result);
                break;

            case '*':
                result = num1 * num2;
                System.out.println("Result: " + result);
                break;

            case '/':
                result = num1 / num2;
                System.out.println("Result: " + result);
                break;

            case '%':
                result = num1 % num2;
                System.out.println("Result: " + result);
                break;

            default:
                break;
        }
        scanner.close();
    }
}
```

```
Enter the First Number: 10
Enter the Second Number: 2
Enter the Operator (+ , - , * , / , %): *
Result: 20.0
```

Q8

```
// Q8
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {

        for(int i = 1; i <= 12; i++) {
            System.out.println(i);
        }
    }
}
```

```
1
2
3
4
5
6
7
8
9
10
11
12
```

Q9

```java
// Q9
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {

        for(int i = 1; i <= 5; i++){
            for(int j = 1; j <= i; j++) {
                System.out.print(j);
            }
            System.out.println();
        }
    }
}
```

```
1
12
123
1234
12345
```

Q10

```java
// Q10
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {

        int i  = 2;
        int sum = 0;

        while (i <= 100) {
            sum = sum + i;
            i = i + 2;
        }

        System.out.println(sum);
    }
}
```

```
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2550
PS D:\NSBM\1 Year
```

Q11

```java
// Q11
import java.util.Scanner;

public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        int num;
        int sum = 0;

        do {
            System.out.print(s: "Enter a number: ");
            num = scanner.nextInt();

            if (num > 0) {
                sum++;
            }

        } while (num > 0);

        System.out.println("Positive numbers recorded: " + sum);

        scanner.close();
    }
}
```

```
ming\Code\User\workspaceStorage\f
Enter a number: 21
Enter a number: 32
Enter a number: 44
Enter a number: 34
Enter a number: -2
Positive numbers recorded: 4
```

Q12

```java
// Q12
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {

        for (int i = 1; i <= 100; i++) {
            if (i == 50) {
                break;
            }
            System.out.print(i + " ");
        }
    }
}
```

```
ming\Code\User\workspaceStorage\f7cf3e8c0fa0aa806e08cd580d2369dc\redhat.java\jdt_ws\1_2e93afdd\bin' 'Assignment1'
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49
PS D:\NSBM\1 Year\2 Sem\Java\Assignment1>
```

Q13

```java
// Q13
public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {

        for (int i = 1; i <= 20; i++) {
            if (i % 3 == 0) {
                continue;
            }
            System.out.print(i + " ");
        }
    }
}
```

```
ming\Code\User\workspaceStorage\f7cf3e8c0fa0a
1 2 4 5 7 8 10 11 13 14 16 17 19 20
PS D:\NSBM\1 Year\2 Sem\Java\Assignment\1>
```

Q14

```java
// Q14
import java.util.Scanner;

public class Assignment1 {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter first number: ");
        int a = sc.nextInt();

        System.out.print(s: "Enter second number: ");
        int b = sc.nextInt();

        System.out.print(s: "Enter third number: ");
        int c = sc.nextInt();

        if (a >= b && a >= c) {
            System.out.println("Largest number: " + a);
        } else if (b >= a && b >= c) {
            System.out.println("Largest number: " + b);
        } else {
            System.out.println("Largest number: " + c);
        }

        sc.close();
    }
}
```

```
ming\Code\User\workspaceStorage\f7cf3e
Enter first number: 10
Enter second number: 23
Enter third number: 43
Largest number: 43
PS D:\NSBM\1 Year\2 Sem\Java\Assignmen
```

Q15

```java
// Q15
import java.util.Scanner;

public class Q15_ATM {
    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int balance = 10000;
        int choice;

        do {
            System.out.println("\nCurrent Balance: " + balance);
            System.out.println(x: "1. Withdraw");
            System.out.println(x: "2. Exit");
            System.out.print(s: "Enter choice: ");
            choice = sc.nextInt();

            if (choice == 1) {
                System.out.print(s: "Enter amount: ");
                int amount = sc.nextInt();

                if (amount <= 0) {
                    System.out.println(x: "Invalid amount!");
                    continue;
                }

                if (amount > balance) {
                    System.out.println(x: "Insufficient balance!");
                    continue;
                }

                balance -= amount;
                System.out.println(x: "Withdraw successful!");

                if (balance < 500) {
                    System.out.println(x: "Balance below 500. Transaction stopped.");
                    break;
                }

            } else if (choice == 2) {
                System.out.println(x: "Thank you!");
                break;
            } else {
                System.out.println(x: "Invalid choice!");
            }

        } while (true);

        sc.close();
    }
}
```

```
Current Balance: 10000
1. Withdraw
2. Exit
Enter choice: 1
Enter amount: 10000
Withdraw successful!
Balance below 500. Transaction stopped.
```